

MONICA (MENGQI) LI

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SUMMARY

- PhD student in Department of Computer engineering, Polytechnique Montréal. Focus on computer vision and robotics.
- Led collaborative projects, resulting in seven peer-reviewed publications, including six first-authored publications
- Deep understanding of machine learning, genomic data analysis and visualization

EDUCATION

Visiting Doctorate Student

Department of Engineering Science Advisor: Daniele De Martini

University of Oxford

2026-current

- Multi-agent map merging for 3D Gaussian reconstruction.
- Using functional map for 3D Gaussian registration.

Ph.D. in Computer Engineering

Making Innovative Space Technology(MIST) Lab Advisor: Prof. Pierre-Yves Lajoie, Prof. Giovanni Beltrame

Polytechnique Montréal

2023 - Current

- Compact optimization on multi-agent 3D reconstruction
- Active view selection on 3D Gaussian Splatting mapping
- Use representative learning to design specific control strategies for multiple robot arms, GPA: 4.0

M.Sc in Computer Science

Advisor: Prof. Madjid Allili

Bishop's University

2020 - 2022

- Research in human motion recognition and computer vision
- Thesis: Human action recognition with feature-embedding graph convolutional network.
- Extra credits in undergraduate computer science and mathematical courses, GPA: 93.6/100

M.A. in Comparative and Public History

Advisor: Prof. Chiu Peng-sheng

Chinese University of Hong Kong

2014 - 2015

- Established a dataset for Ba County Archives (General Research Fund #14402414)

B.B.A IN ACCOUNTANCY

The Hong Kong Polytechnic University

2009 - 2013

- Mass data collection from SEC database for Prof. Simon Fung Yu Kit's accounting research in labor union
- Exchange program to Anderson School of Management, University of New Mexico in 2011

ACADEMIA EXPERIENCE

Bishop's University - Department of Computer Science

Teaching Assistant

Sherbrooke, QC, Canada

Feb 2021 - Aug 2022

- Mark students' assignments and tests for the course "Foundations Computer Science" and "Medical Image".
- Answer students' questions related to the lecture materials.
- Develop a SQL parser that provides detailed feedback (beyond the traditional syntax error) on the student work in an English/French readable form. The parser support grading and detailed feedback.
- Test the SQL parser with a standardised set of SQL queries.

City University of Hong Kong

Supervisor: Dr. Howard Leung

3D Motion Capture Laboratory

Research Assistant

May 2018 to Sep 2020

- Adopted graph centrality and motif to augment the original graph convolutional network (GCN) for motion classification tasks.
- Developed an algorithm to quantify features of online Chinese handwriting, including slant/tilt, horizontal/vertical alignment, deviation of size and pressure.
- Utilised SPSS for traditional statistics analysis.
- Recorded 20 subject EEG data for handwriting ability assessment and understanding.
- Ad-hoc consultancy for two individual undergraduate final year projects and three group final year projects.
- These projects led to the submission of 4 publications.

PUBLICATIONS

Li, M. M., Lajoie, P.Y., Beltrame, G. *Compact Keyframe-Optimized Multi-Agent Gaussian Splatting SLAM*. In 2026 IEEE International Conference on Robotics & Automation (ICRA) MM-SpatialAI Workshop doi.org/10.48550/arXiv.2409.16470

Li, M. M., Lajoie, P.Y., Beltrame, G. *Frequency-based View Selection in Gaussian Splatting Reconstruction*. In 2026 Conference on Robots and Vision(CRV) doi.org/10.48550/arXiv.2409.16470

Li, M. M., *A Motion Capture Quality Comparison Between Rokoko and Kinect*. In 2024 IEEE International Conference on Artificial Intelligence in Engineering and Technology(IICAET) (pp. 239-242). IEEE.[10.1109/IICAET62352.2024.10730257](https://doi.org/10.1109/IICAET62352.2024.10730257)

Li, M. M., *Exploration of the Possibility for a Wider Range of the Discount Factor for Reinforcement Learning*, 2024 6th International Conference on Communications, Information System and Computer Engineering, pp. 1536-1539. doi.org/10.1109/CISCE62493.2024.10653102

Li, M. M., Belzile, B., Imran, A., Birglen, L., Beltrame, G., St-Onge, D. . *From Assistive Devices to Manufacturing Cobot Swarms*, 32nd IEEE International Conference on Robot and Human Interactive Communication(RO-MAN) doi.org/10.1109/RO-MAN57019.2023.10309449

Zhang, W., Ao, Q., Guan, Y., Zhu, Z., Kuang, D., Li, M., ... , Shen, K. (2021). *A Novel Diagnostic Approach For The Classification Of Small B-Cell Lymphoid Neoplasms Based On Nanostring Platform*, Modern Pathology - Nature <https://www.nature.com/articles/s41379-021-00954-z>

Li M, Leung H, Li TMH, Li-Tsang CWP *Measuring the tilt and slant of Chinese handwriting in primary school students: A computerized approach*, PLOS ONE 14(11): eo223485. <https://doi.org/10.1371/journal.pone.0223485>

Ma L and Li M, *A Quantitative Way to Utilise the Social Network in Social Status: A Study Using CBDB Song Dynasty Data*, presented at the DADH 2019: 10th International Conference of Digital Archives and Digital Humanities. <https://dadh2019.conf.tw/site/page.aspx?pid=305&sid=1308&lang=en#paper7>

INDUSTRIAL EXPERIENCE

Pratt and Whitney Canada - Technology Collaborative Office

Longueuil, QC, Canada

Internship - Collaborative research projects coordination & technology management process improvement May 2023 - Sep 2023

- Design compliance and internal control procedures for intellectual property transfer.
- Contacted the balloters for the procedures.
- Deploy the control procedures to corporate WEBCON.

Aeroport AI

Montreal, QC, Canada

Machine Learning Consultant

Jan 2023 - Apr 2023

- Trained deep learning models for anomaly detection.
- Deployed the model onto AWS server.
- Developed the SDK and user interface for collaborators to use the model.

PricewaterhouseCoopers Hong Kong

Assurance - Financial Service, Hong Kong Office

Associate Auditor

Sep 2015 - Jun 2017

- Work on several annual auditing projects for banks, insurance companies and mutual funds.
- Conduct an IPO audit for a listed company.
- Perform compliance regulation checks for two investment banks.

Ernst & Young Hua Ming LLP Shanghai Branch

Shanghai, China

Internship - Assurance

Aug 2014

- Bank and financial lease confirmation work to ensure an airline company's debt status.
- Translate letters of representation from English into Chinese as the client required.
- Industrial research data collection.

AWARDS

- Vanier Canada Graduate Scholarships (Vanier CGS) - Natural Sciences and Engineering Research Council (\$50,000 × 3 years)
- Richard J. Schmeelk Canada Fellowship (\$10,000 × 4 years)
- Mitacs Globalink Research Award Abroad (\$6,000)
- Canada Graduate Scholarships - Michael Smith Foreign Study Supplements (\$6,000)
- CRIAQ Next Generation of Women in Aerospace (\$15,000)
- WEDO Canada National scholarship (\$2,500)
- AQEIPS bursary program (\$2,000)
- Amelia Earhart Fellowship (\$10,000 USD)

VOLUNTEERING AND OUTREACH SERVICES

Olympes de la Parole Canada

Nov 2024 - Jan 2026

- Mentoring two teams of high school girls to study gender gap in STEM fields.
- Provide opinions for research proposal and experiment design of their projects.
- The research projects were presented to the Commission on the Status of Women at United Nations.

Bishop's University

Residence Life

Resident Assistant

Sep 2021 - Dec 2021

- Worked on residence life activities and community preparation.
- Facilitate students to live in the residence and respond to their needs.
- Conduct duty shifts in the residential halls at night.